

HIGH-INTENSITY INTERMITTENT TRAINING IN HYPOXIA: A DOUBLE-BLINDED, PLACEBO-CONTROLLED FIELD STUDY IN YOUTH FOOTBALL PLAYERS

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ABSTRACT

Brocherie, F, Girard, O, Faiss, R, and Millet, GP. High-intensity intermittent training in hypoxia: A double-blinded, placebo-controlled field study in youth football players. *J Strength Cond Res* 29(1): 226–237, 2015—This study examined the effects of 5 weeks (~60 minutes per training, 2 d·wk⁻¹) of run-based high-intensity repeated-sprint ability (RSA) and explosive strength/agility/sprint training in either normobaric hypoxia repeated sprints in hypoxia (RSH; inspired oxygen fraction [F_IO₂] = 14.3%) or repeated sprints in normoxia (RSN; F_IO₂ = 21.0%) on physical performance in 16 highly trained, under-18 male footballers. For both RSH (*n* = 8) and RSN (*n* = 8) groups, lower-limb explosive power, sprinting (10–40 m) times, maximal aerobic speed, repeated-sprint (10 × 30 m, 30-s rest) and repeated-agility (RA) (6 × 20 m, 30-s rest) abilities were evaluated in normoxia before and after supervised training. Lower-limb explosive power (+6.5 ± 1.9% vs. +5.0 ± 7.6% for RSH and RSN, respectively; both *p* < 0.001) and performance during maximal sprinting increased (from -6.6 ± 2.2% vs. -4.3 ± 2.6% at 10 m to -1.7 ± 1.7% vs. -1.3 ± 2.3% at 40 m for RSH and RSN, respectively; *p* values ranging from <0.05 to <0.01) to a similar extent in RSH and RSN. Both groups improved best (-3.0 ± 1.7% vs. -2.3 ± 1.8%; both *p* ≤ 0.05) and mean (-3.2 ± 1.7%, *p* < 0.01 vs. -1.9 ± 2.6%, *p* ≤ 0.05 for RSH and RSN, respectively) repeated-sprint times, whereas sprint decrement did not change. Significant interactions effects (*p* ≤ 0.05) between condition and time were found for RA ability-related parameters with very likely greater gains (*p* ≤ 0.05) for RSH than RSN (initial sprint: 4.4 ± 1.9% vs. 2.0 ± 1.7% and cumulated times: 4.3 ± 0.6% vs. 2.4 ± 1.7%). Maximal

aerobic speed remained unchanged throughout the protocol. In youth highly trained football players, the addition of 10 repeated-sprint training sessions performed in hypoxia vs. normoxia to their regular football practice over a 5-week in-season period was more efficient at enhancing RA ability (including direction changes), whereas it had no additional effect on improvements in lower-limb explosive power, maximal sprinting, and RSA performance.

KEY WORDS normobaric hypoxia, hypoxic training, repeated-sprint ability, agility, football (soccer)

INTRODUCTION

In football (soccer), players must possess well-developed technical and tactical skills and a high fitness to cope with the physical demands of the game at the professional level (54). Despite a large number of tests and the growing scientific debate regarding the tradeoffs between laboratory and field evaluation, valid and reliable field tests (50) are usually preferred for performance evaluation to better reflect football specificity (54). Such field tests should include an evaluation of aerobic endurance, muscular strength, peak speed and power, acceleration, repeated-sprint ability (RSA), and the ability to change direction (COD) to reflect the complexity of the game. Crucial determinants of match-winning situations include short-sprint accelerations, jumps, and football-specific skills (e.g., dribbling, short-passing ability, shooting), requiring well-developed muscle strength capacity and neuromuscular control strategies (43,49).

To enhance these run-based abilities, diverse training strategies focusing on important physical aspects of the game (i.e., aerobic, sprint, repeated sprint and resistance training) have been proposed (3,6). Considering the physiological requirements and high energetic demands of football match play (i.e., sustained [90 minutes] high-intensity intermittent activity with mean and peak heart rates of approximately 85 and 98% of maximal values) (28), training modalities that specifically enhance the capacity to repeat

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TABLE 1. Subjects' anthropometric characteristics.*†

	RSN (<i>n</i> = 8)	RSH (<i>n</i> = 8)	RSH-RSN (Cohen's <i>d</i> ± 95% CL)
Age, y	17.1 ± 0.2 (16.7–17.5)	17.1 ± 0.3 (16.6–17.4)	0.26 ± 1.21
Weight, kg	56.1 ± 9.1 (45.2–69.1)	62.4 ± 6.1 (51.7–68.2)	0.62 ± 1.66
Height, cm	170.0 ± 5.5 (162.8–177.6)	172.5 ± 3.0 (168.7–175.7)	0.69 ± 1.11
BMI, kg·m ⁻²	19.3 ± 2.4 (16.1–21.9)	21.0 ± 1.7 (18.2–22.8)	0.47 ± 1.89
Sum of 7 skinfolds, mm	41.4 ± 4.6 (35.4–48.7)	48.2 ± 13.7 (31.3–72.9)	0.87 ± 2.18

*RSN = repeated-sprint in normoxia; RSH = repeated-sprint in hypoxia; CL = confidence limits.

†Values are given as mean ± SD (range), between-condition differences (Cohen's *d* ± 95% CL).

high-intensity efforts have clear benefits for these athletes (12,53), with a likely direct transfer to the playing field (28). For instance, both high-intensity intermittent exercise and RSA are complex fitness components that depend on both muscular (e.g., oxidative capacity, phosphocreatine recovery, and H⁺ buffering) and neural (e.g., muscle activation and recruitment strategies) factors among others. Although the best training methods to improve RSA are widely debated (3,6), the additional use of hypoxic stress in sport-specific test settings has recently been introduced to the discussion (2,39) with the emergence of new promising training strategies including repeated sprints in hypoxia (RSH) (20,22,40,47). Beside the wide range of exercise prescription and denomination beyond the scope of high-intensity intermittent training (55) and according to the recently updated nomenclature for the different hypoxic

methods available for team sports (40), we would preferably use the terms related to RSA to define the training prescribed in this study.

Repeated sprints in hypoxia are based on the repetition of “all-out” efforts of short (<30 seconds) duration interspersed with short incomplete recoveries under hypoxic conditions (40). It differs from intermittent hypoxic training because the intensity of the training stimulus is maximal and presumably allows for maintaining high fast-twitch recruitment so that positive results when adding hypoxia to training can be expected.

A RSH training stimulus is particularly interesting because under hypoxic conditions (up to 3,800 m) single-sprint performance of short duration (<10 seconds) is generally preserved, whereas fatigue resistance during RSA tests is compromised with earlier and larger decrements in mechanical work (4). For example, Bowtell et al. (4) has recently demonstrated that

TABLE 2. Training content during a typical training week.*†

Day	1	2	3	4	5	6	7
Morning							
Activity	Football (skills + tactics)	Football (skills + tactics)	Football (skills + tactics)	Football (tactics)	Recovery/skills	Football (agility + skills)	Off
Duration, min	60	60	60	30	45	30	
Intensity, %	~60	~60	~70	~50	~50	~70	
HR _{max}							
Afternoon							
Activity	Conditioning	Football (skills + tactics)	Conditioning	Friendly match (or training)	Football (skills + tactics)	Domestic match	Off
Duration, min	60	75	60	90	75	90	
Intensity, %	~90–100	~70	~90–100	~80	~70	~85–90	
HR _{max}							

*% HR_{max} = % of maximal heart rate.

†Cells in bold refer to activities performed in hypoxic environment.

during nonmotorized treadmill repeated-sprint runs (10×6 -s sprints with 30-second rest), the peak running speed was relative resilient to hypoxic exposure (inspired oxygen fraction [$F_{I}O_2$] ranging from 12 to 15%).

Compared with similar training at sea level, RSH seems to induce larger benefits on glycolytic pathways, skeletal muscle adaptation, and ventilatory responses, which suggests putative stronger benefits for team sport physical performance (19). For instance, RSH was shown to delay fatigue during an RSA cycling test (i.e., 10-s sprints interspersed with 20-s recoveries) to exhaustion by 40% compared with the same training performed (repeated sprints) in normoxia (RSN) (20). An additional benefit of RSH vs. RSN was also found by Galvin et al. (22) with well-trained academy rugby union and rugby league players' intermittent running performance (i.e., Yo-Yo Intermittent Recovery Test level 1) being improved by 33% after 4 weeks of training including 120 running sprints of 6 seconds. However, with only 1 study including team sport athletes (22), further studies are still needed to endorse the efficacy of RSH in football players, notably by testing more ecological game situations including agility tests and overground (repeated) sprints (19).

Using a randomized, double-blinded placebo-controlled design, this study aimed at comparing the effects of run-based RSH vs. RSN, combined with sea level football-specific training on several physical fitness parameters in highly trained youth male football players. We hypothesized that RSH would improve neuromuscular factors (i.e., lower-limb explosive power and sprinting abilities) and specific repeated-sprint abilities performance to a greater extent than the equivalent training in normoxia (RSN).

METHODS

Experimental Approach to the Problem

The study was designed as a randomized, balanced, double-blinded placebo-controlled trial, in which the experimental intervention consisted of an additional RSH ($F_{I}O_2 = 14.3\%$) vs. RSN ($F_{I}O_2 = 21.0\%$) training to the usual in-season sea-level football training routine for 5 weeks (~ 60 minutes/training, $2 \text{ d} \cdot \text{wk}^{-1}$) at spring 2008.

After baseline, including 2 familiarization training sessions performed once during the week preceding the supervised training period and premeasurements (Pre-), all participants were randomly assigned to 1 of the 2 training groups based on their age and physical performance results. Training content included high-intensity intermittent and RSA exercises and explosive strength/agility tasks either in hypoxia (RSH, $n = 8$) or normoxia (placebo group, RSN = 8) in addition to their usual football-specific training at sea level. Posttesting (Post-) session was repeated the week after the 5-week supervised specific training period (10 training sessions). Both Pre- and Post- tests were performed in an invariant sequence (on 2 separate occasions; day 1: lower-limb explosive power, linear sprint, RSA and repeated-agility [RA] tests; and day 2: incremental field running test to

estimate maximal aerobic speed [MAS] with no more than 4 days between each test session) in normoxia on an indoor track at constant temperature of $22.0 \pm 0.5^\circ \text{C}$ and $55 \pm 10\%$ of relative humidity. Players were told not to perform any intense exercise on the day before the test and to consume their last meal at least 3 hours before the scheduled test time. All players were familiar with all testing procedures and exercises, as part of their usual battery of team fitness testing and training routines. None of the subjects had any previous experience of altitude exposure. All of them were told that training was to be performed in hypoxia.

Subjects

The sample size was estimated using acceptable precision or confidence intervals (CI) a priori using the approach developed for magnitude-based inferences (27). Based on the assumption that a between-group difference in mean RSA time of $1.2 \pm 1.1\%$ is meaningful (11,13) and considering a within-subject *SD* (typical error) of 0.8% (29), a sample size of >7 participants per group would provide maximal chances of 0.5 and 25% of type I and type II errors, respectively. Thus, 16 highly trained under-18 male football players participated in this study. This research project was approved by the local research ethics committee, and conformed to the recommendations of the Declaration of Helsinki for use of Human Subjects as per the *Journal of Strength and Conditioning Research* author guidelines. The players and their parents were provided with the procedures and risks associated with participation in the study. Written informed consent was obtained from the players and their parents (for minors). Only outfield players were included in the present study. Subjects' anthropometric characteristics are shown in Table 1.

All the players had more than 4 years training history in a high-performance football academy and participated on average in 12 hours of combined football training and competitive games per week (6–8 football training sessions, 2 conditioning sessions, 1 domestic match each week and 1–2 friendly international club matches every 3 weeks). Football training sessions lasted 60–90 minutes and were dedicated to improve technical skills and tactical attributes. Such work can be qualified as moderate-intensity intermittent aerobic exercise (55–85% of maximal heart rate, HR_{max}) with regular occurrence of short sprints (12). Matches can be regarded as high-intensity intermittent exercise, with young players likely to spend about half of their playing time at intensity $>85\%$ of HR_{max} (25). Two training sessions (morning and afternoon training sessions or morning training + afternoon match) were programmed on most days. Recovery sessions were planned the day after a match. A typical weekly training content during the length of the study is presented in Table 2.

Procedures

Training Intervention. All players followed their usual football-specific training sessions during the protocol, though excluding sprinting and explosive exercises. Additionally, all groups performed specific run-based training sessions in

TABLE 3. Outline of the training intervention for the high-intensity training in normoxia (RSN) and hypoxia (RSH) groups.*

High-intensity intermittent treadmill runs							
Week	Interval, s	Sets × reps	Intensity (% MAS)	Rest between sets, min	Rest type	SpO ₂ , %	% HRmax
1	15:15	2 × 5	90	5 (except for the last set recovery period: 10 min)	Light jogging	RSN, 98 ± 2 RSH, 91 ± 2	95 ± 3 97 ± 2
2	15:15	2 × 6	100		Technical skills	RSN, 97 ± 1 RSH, 91 ± 3	97 ± 2 98 ± 1
3	15:15	3 × 7	100		Light jogging	RSN, 97 ± 1 RSH, 92 ± 1	96 ± 2 98 ± 1
4	15:15	3 × 8	110		Technical skills	RSN, 97 ± 2 RSH, 91 ± 2	98 ± 1 99 ± 1
5	15:15	3 × 9	110		Technical skills	RSN, 97 ± 2 RSH, 92 ± 3	98 ± 1 99 ± 1
Repeated-sprint exercises†							
Week	Plyometric jumps (CMJs to box) Sets × reps (height, cm)	Agility drills (ladders) (ladders) Sets × reps	Plyometric jumps (hurdles) Sets × reps	Acceleration (standing starts) Set × reps (d in m)	Repeated shuttle sprints (<5 s COD) Set × reps (d in m; t in s)	SpO ₂ , %	% HRmax
1	4 × 4 (30)	4 × 4	4 × 4	4 × 4 (5)	4 × 4 (10; 15)	RSN, 97 ± 1 RSH, 93 ± 1	84 ± 3 90 ± 3
2	5 × 3 (30)	5 × 3	5 × 3	5 × 3 (5)	5 × 3 (10; 15)	RSN, 97 ± 1 RSH, 94 ± 1	83 ± 3 88 ± 3
3	5 × 4 (45)	5 × 4	5 × 4	5 × 4 (10)	5 × 4 (5; 15)	RSN, 97 ± 1 RSH, 93 ± 1	83 ± 2 87 ± 3
4	6 × 3 (30)	6 × 3	6 × 3	6 × 3 (5)	6 × 3 (10; 15)	RSN, 98 ± 1 RSH, 94 ± 1	82 ± 4 88 ± 1
5	6 × 4 (45)	6 × 4	6 × 4	6 × 4 (10)	6 × 4 (5; 15)	RSN, 98 ± 1 RSH, 94 ± 1	81 ± 4 87 ± 2

*RSN = repeated-sprint in normoxia; RSH = repeated-sprint in hypoxia; MAS = maximal aerobic speed; SpO₂ = arterial oxyhemoglobin saturation; % HRmax = % of maximal heart rate; CMJs = countermovement jumps; COD = change of direction; d = distance of run; t = duration of passive recovery.

†All performed at maximal speed/sense of effort.

a normobaric hypoxic room (30 × 30 m; Colorado altitude training, Louisville, CO, USA), where temperature and relative humidity were kept constant (22.0 ± 0.5° C and 55 ± 10%, respectively). The inspired oxygen fraction (F_IO₂) was 14.3% to simulate an altitude of 2,900 m. To blind subjects to altitude, the system was also run for RSN with a normoxic airflow into the chamber. During all training sessions, fingertip pulse oximeters (Wristox 3100; Nonin, Plymouth, MN, USA) were used to estimate arterial oxyhemoglobin saturation (SpO₂), which was recorded at rest and immediately after every sprint (5 seconds) for each experimental trial, whereas only session averaged values are considered for further analysis. Participants were unable to view any heart rate and SpO₂ readings. Furthermore, all investigators, except for the main investigator (G.P.M.), were blinded toward the group assignment.

After a standardized warm-up (~15 minutes) in the normobaric hypoxic room, each RSH or RSN training session (~45 minutes) included:

- high-intensity (ranging from 90 to 110% of MAS or 16 km·h⁻¹ < MAS < 20 km·h⁻¹) intermittent exercises consisting of 2–3 sets of 5–6 running bouts of durations ~15 seconds performed on a motorized treadmill (Woodway PPS Med; Woodway, Waukesha, WI, USA) interspersed with 15 seconds of passive recovery as adapted from previous studies (12,53). Recovery between the sets (5 and 10 minutes) involved either light running or combined with easy technical exercises inside the room; and
- repeated-sprint exercise including explosive strength/COD/sprint drills consisting of 4–6 series of 3–4 repetitions performed in the training room, aiming exclusively at improving speed, acceleration, and agility

TABLE 4. Mean changes in lower-limb explosive power, sprinting times, MSS, and MAS after repeated-sprint training performed either in normoxia (RSN) or hypoxia (RSH).^{*†‡}

		Pre-	Post-	Post- Pre- (Cohen's <i>d</i>)	Post- Pre- qualitative inference	<i>p</i>
CMJ, cm	RSN	41.5 ± 4.6	43.8 ± 3.8	0.54 ± 1.57	Possibly beneficial	<0.001
	RSH	40.3 ± 3.2	42.9 ± 3.1	0.70 ± 0.18	Most likely beneficial	<0.001
	RSH-RSN (Cohen's <i>d</i>)	-0.29 ± 1.15				NS
	Qualitative inference	Possibly beneficial				
10 m, s	RSN	1.78 ± 0.06	1.70 ± 0.03	-1.01 ± 2.36	Likely beneficial	<0.05
	RSH	1.84 ± 0.05	1.73 ± 0.05	-1.97 ± 0.67	Most likely beneficial	<0.01
	RSH-RSN (Cohen's <i>d</i>)	0.68 ± 1.17				NS
	Qualitative inference	Likely beneficial				
20 m, s	RSN	3.04 ± 0.08	2.95 ± 0.06	-0.46 ± 2.30	Possibly beneficial	<0.01
	RSH	3.08 ± 0.07	2.98 ± 0.08	-1.11 ± 0.60	Most likely beneficial	<0.01
	RSH-RSN (Cohen's <i>d</i>)	0.15 ± 1.17				NS
	Qualitative inference	Possibly beneficial				
30 m, s	RSN	4.22 ± 0.11	4.15 ± 0.10	-0.02 ± 2.46	Possibly beneficial	<0.01
	RSH	4.25 ± 0.11	4.15 ± 0.11	-0.80 ± 0.57	Very likely beneficial	<0.01
	RSH-RSN (Cohen's <i>d</i>)	0.33 ± 1.17				NS
	Qualitative inference	Possibly beneficial				
40 m, s	RSN	5.39 ± 0.13	5.32 ± 0.14	0.15 ± 2.37	Possibly beneficial	NS
	RSH	5.40 ± 0.13	5.31 ± 0.15	-0.58 ± 0.55	Likely beneficial	<0.05
	RSH-RSN (Cohen's <i>d</i>)	0.19 ± 1.17				NS
	Qualitative inference	Possibly beneficial				
MSS, km·h ⁻¹	RSN	31.0 ± 1.0	30.7 ± 1.1	-0.73 ± 2.25	Possibly trivial	NS
	RSH	31.2 ± 0.8	31.1 ± 0.9	-0.15 ± 0.48	Possibly trivial	NS
	RSH-RSN (Cohen's <i>d</i>)	-0.16 ± 1.19				NS
	Qualitative inference	Possibly trivial				
MAS, km·h ⁻¹	RSN	17.8 ± 1.3	17.8 ± 1.56	-0.77 ± 2.60	Very unlikely beneficial	NS
	RSH	17.3 ± 0.7	17.4 ± 0.6	0.22 ± 0.51	Possibly beneficial	NS
	RSH-RSN (Cohen's <i>d</i>)	0.55 ± 1.32				NS
	Qualitative inference	Possibly beneficial				

*Pre- = before training; Post- = after 5-week experimental period; CMJ = countermovement jump; RSN = repeated-sprint in normoxia; RSH = repeated-sprint in hypoxia; NS = not significant; MSS = maximal sprint speed; MAS = maximal aerobic speed; CL = confidence limits.

†Mean ± SD, within-condition changes (Cohen's *d* ± 95% CL) and between-condition differences in the change (Cohen's *d* ± 95% CL) in for sprinting times at 10, 20, 30, 40 m and MSS Pre- and Post- test.

‡Qualitative inference: % chances for changes in RSH to be greater/similar/smaller than those observed in RSN.

(e.g., plyometric, agility drills, standing start and very short shuttle sprints, all performed at maximal speed/sense of effort, <5 seconds in duration or ≤10 m distances) as previously described (12); repetitions and series were interspersed with at least 45 seconds and 3 minutes of passive recovery, respectively.

These specific training sessions were completed at the same time of day (±2 hours) and were practiced 2 times per week (i.e., once for each session) for a total of 10 sessions during the experiment. Table 3 shows the details of the training contents/intensities during the experimental period.

Lower-Limb Explosive Power Test. Countermovement jump (CMJ, cm) height was determined as the center of mass vertical displacement calculated from body mass-corrected force development (Bioware software version 4.0; Kistler Instrument Corp., Winterthur, Switzerland)

from a force platform (Kistler 9286AA; Kistler Instrument Corp., Winterthur, Switzerland). Each trial was validated by visual inspection to ensure each landing was without any leg flexion and participants were instructed to keep their hands on their hips during all jumps. The CMJ was performed 3 times, separated by 45 seconds of passive recovery, and the highest jump was recorded. Among all popular jumping tests, CMJ was previously demonstrated as the most reliable and valid jumping test (coefficient of variation [CV] = 2.8%) for the estimation of explosive power of the lower limbs in physically active collegial men (19.6 ± 2.1 years) (34).

Single sprints. Players were asked to run 2 maximal, straight-line 40-m sprints during which 10-m split times were recorded. Sprinting time was measured to the nearest 0.01 seconds using dual-beam electronic timing gates (Swift

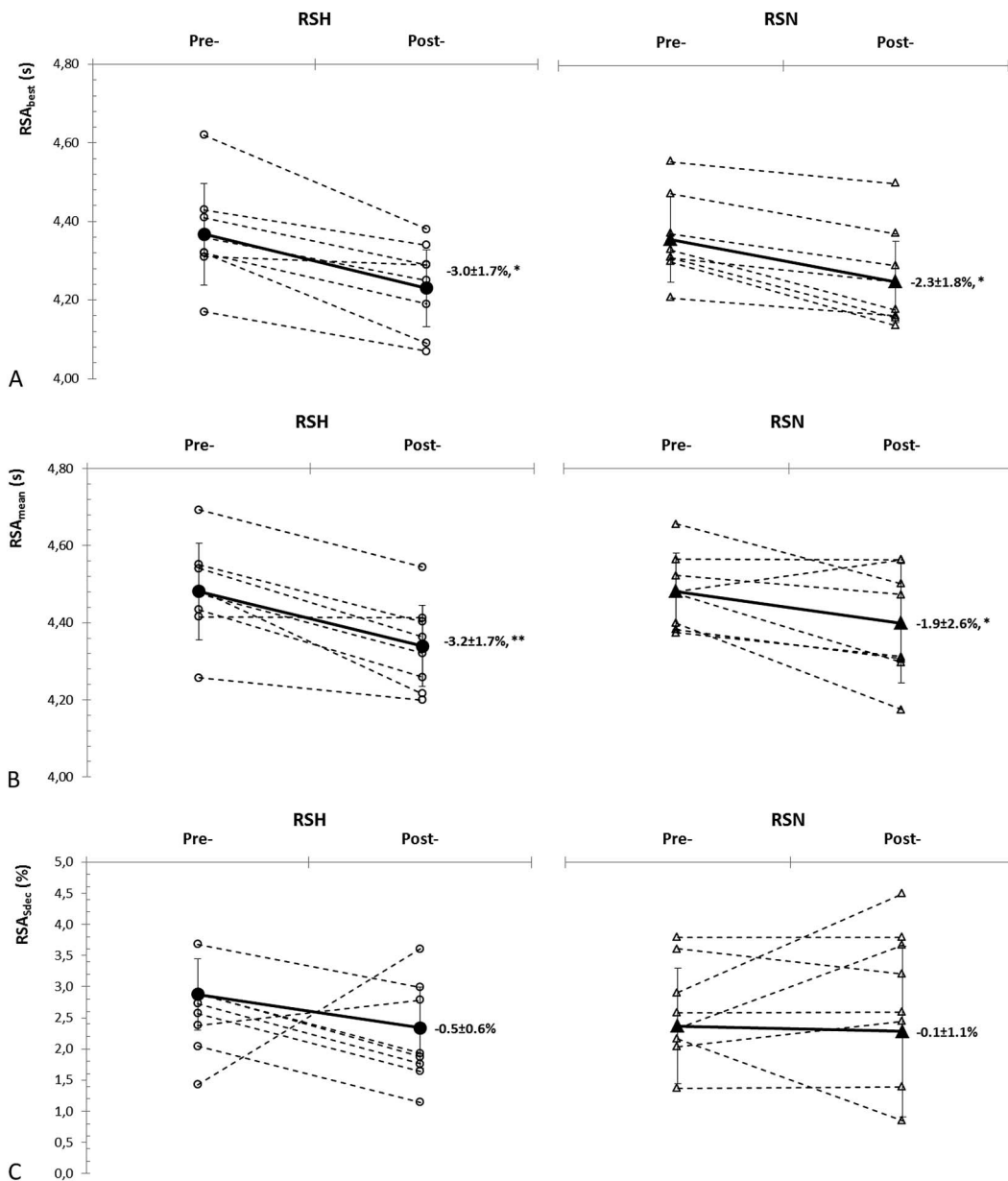


Figure 1. Individual (small symbols, dotted thin lines) and average (large symbols, dark line) best (RSA_{best}; A), mean (RSA_{mean}; B) RSA and sprint decrement (RSA_{sdec}; C) before (Pre-) and after (Post-) repeated-sprint training in hypoxia (RSH; ○) or normoxia (RSN; Δ). Values are mean ± SD. Significant differences from pretest, * $p \leq 0.05$ and ** $p < 0.01$, respectively.

Performance Equipment, Lismore, Australia). The height of the photocell was adjusted according to the height of the participant's hip. Players started sprint when ready (thus eliminating reaction time) from a standing static position with their front foot 0.5 m behind the first timing gate and were instructed to sprint as fast as possible over the sprint distance and the best performance was kept for analysis. Players' maximal sprint speed ($\text{km} \cdot \text{h}^{-1}$) was defined as the average running speed attained during the

fastest 10-m split. The reliability of maximal sprinting speed was assessed in a group of young soccer players: the typical error, expressed as CV was 1.4% (7).

Repeated-Sprint Ability and Repeated-Agility Tests. The RSA test involved 10 repetitions of straight-line maximal 30 m in alternating directions interspersed by 30 seconds of passive recovery. A relatively similar RSA field test (i.e., 7×30 m with 25 seconds of active recovery) was found reliable

TABLE 5. Mean changes in RSAs after repeated-sprint training performed either in normoxia (RSN) or hypoxia (RSH). *†‡

		Pre-	Post-	Post- Pre- (Cohen's <i>d</i>)	Post- Pre- qualitative inference	<i>p</i>
RSA _{best} , s	RSN	4.37 ± 0.11	4.26 ± 0.10	-1.48 ± 1.61	Very likely beneficial	<0.05
	RSH	4.37 ± 0.13	4.23 ± 0.10	-1.04 ± 0.55	Very likely beneficial	<0.05
	RSH-RSN (Cohen's <i>d</i>)	0.38 ± 1.24				NS
	Qualitative inference	Possibly beneficial				
RSA _{mean} , s	RSN	4.49 ± 0.10	4.42 ± 0.15	-1.07 ± 1.12	Very likely beneficial	<0.05
	RSH	4.48 ± 0.13	4.35 ± 0.11	-1.10 ± 0.55	Most likely beneficial	<0.05
	RSH-RSN (Cohen's <i>d</i>)	0.55 ± 1.32				NS
	Qualitative inference	Possibly beneficial				
RSA _{Sdec} , %	RSN	2.46 ± 0.89	2.38 ± 1.32	-0.08 ± 1.06	Possibly beneficial	NS
	RSH	2.88 ± 0.57	2.34 ± 0.65	-0.54 ± 0.64	Likely beneficial	NS
	RSH-RSN (Cohen's <i>d</i>)	-0.44 ± 1.27				NS
	Qualitative inference	Likely beneficial				
RA _{best} , s	RSN	5.87 ± 0.10	5.76 ± 0.11	-0.99 ± 0.72	Very likely beneficial	<0.05
	RSH	6.03 ± 0.19	5.78 ± 0.11	-1.51 ± 0.63	Most likely beneficial	<0.001
	RSH-RSN (Cohen's <i>d</i>)	1.24 ± 1.07				<0.05
	Qualitative inference	Very likely beneficial				
RA _{mean} , s	RSN	6.00 ± 0.10	5.85 ± 0.14	-1.08 ± 0.65	Very likely beneficial	<0.05
	RSH	6.14 ± 0.16	5.90 ± 0.14	-1.52 ± 0.23	Most likely beneficial	<0.001
	RSH-RSN (Cohen's <i>d</i>)	1.34 ± 1.05				<0.05
	Qualitative inference	Very likely beneficial				
RA _{Sdec} , %	RSN	2.31 ± 0.52	1.61 ± 1.17	-0.69 ± 0.89	Likely beneficial	NS
	RSH	2.14 ± 1.23	2.06 ± 1.01	-0.08 ± 1.92	Possibly beneficial	NS
	RSH-RSN (Cohen's <i>d</i>)	0.35 ± 1.12				NS
	Qualitative inference	Possibly beneficial				

*Pre- = before training; Post- = after 5-week experimental period; RSA = repeated-sprint ability; RSN = repeated-sprint in normoxia; RSH = repeated-sprint in hypoxia; NS = not significant; RA = repeated agility; CL = confidence limits.

†Mean ± SD, within-condition changes (Cohen's *d* ± 95% CL) and between-condition differences in the change (Cohen's *d* ± 95% CL) in for RSA and RA-related parameters at Pre- and Post- test.

‡Qualitative inference: % chances for changes in RSH to be greater/similar/smaller than those observed in RSN.

(CV <2.7% for sprinting times) in young team sports athletes (45). The RA test consisted of 6 × 20 m maximal sprints with COD departing every 30 seconds (adapted from a previous running agility test that has been shown to be reliable and valid; intraclass correlation coefficient [ICC] = 0.90%, technical error of measurement [TEM] = 2.8%) in assessing agility (21). Three cones, ~1 m in height, were placed 5 m apart in the shape of an "L." Players ran forward 5 m, turned 90° to their left, ran forward 5 m, turned 180° on their left, and followed the same course to return to the finish line.

Both RSA and RA abilities were assessed using the aforementioned dual-beam electronic timing gates (see Single sprints section). After deceleration, participants walked back to the starting line and assumed a standing static position with their front foot 0.5 m behind the first timing gate. Five seconds before starting each sprint, the subjects were asked to assume the ready position as detailed for the sprint tests and await the start signal (i.e., announcement of a countdown before each sprint: "5 seconds, 3-2-1, Go"). Strong verbal encouragement was provided to each subject during all sprints. The best sprint time (RSA_{best} and RA_{best}; seconds), usually the first repetition,

was retained, whereas mean repeated-sprint times (RSA_{mean} and RA_{mean}; seconds) and the sprint decrement score (RSA_{Sdec} and RA_{Sdec}; %) ($100 - [(best\ sprint\ time \times number\ of\ sprints) / total\ sprint\ times] \times 100$) were calculated (23).

Incremental Field Test. To estimate MAS, an incremental field test, a modified version of the University of Montreal Track Test (i.e., the VAMEVAL maximal incremental running test (31)) was performed. The VAMEVAL test began with an initial running speed of 8.5 km·hour⁻¹ with a consecutive speed increase of 0.5 km·hour⁻¹ each minute until exhaustion. The players adjusted their running speed according to auditory signals timed to match 20-m intervals delineated by cones around a 200-m-long indoor athletic track. The test ended when participants failed on 3 occasions to reach the next cone in the required time. The average velocity of the last 1-minute stage completed was retained as the player's MAS (km·h⁻¹). If the last stage was not fully completed, the MAS was calculated using the formula of Kuipers et al. (30). Whereas the University of Montreal Track Test, which is very similar to the VAMEVAL, was found reliable ($r =$

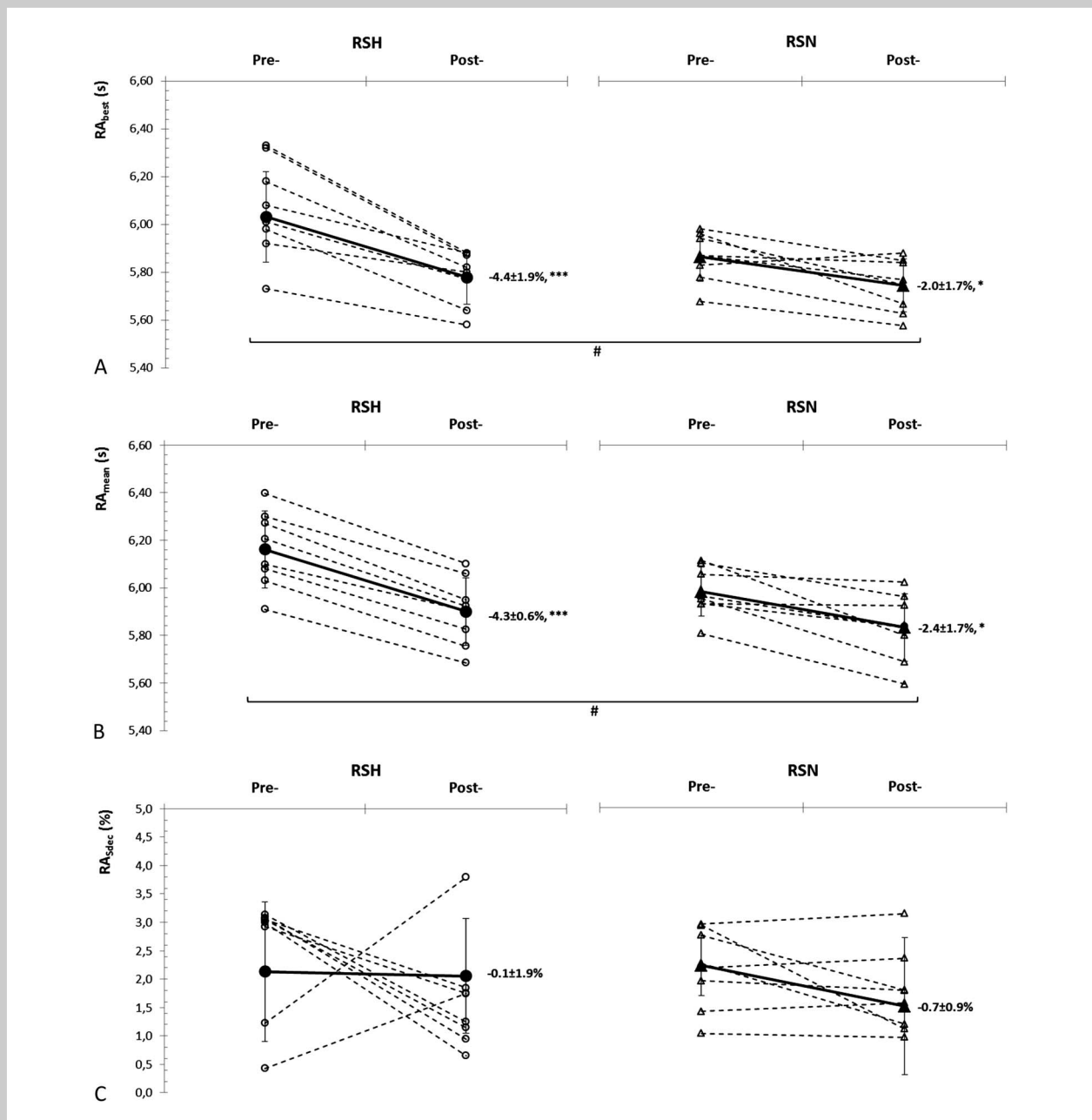


Figure 2. Individual (small symbols, dotted thin lines) and average (large symbols, dark line) in best (RA_{best}; A), mean (RA_{mean}; B) repeated-agility abilities and sprint decrement (RA_{Sdec}; C) before (Pre-) and after (Post-) repeated-sprint training in hypoxia (RSH; ○) or normoxia (RSN; Δ). Values are mean ± SD. Significant differences from Pre- test, **p* ≤ 0.05; and ****p* < 0.001; significant differences between groups, #*p* ≤ 0.05.

0.97) (31), the reliability of VAMEVAL to assess MAS was previously reported in a cohort of high-level academy players. The typical error, expressed as a CV, was 3.5% (14).

Statistical Analyses

Data are presented as mean or relative changes (%) with SD (±SD) unless otherwise stated. Normal distribution of the

data was tested using the Shapiro-Wilk test. Data were first analyzed using a 2-factor repeated-measure analysis of variance with 1 between factor (condition; RSH vs. RSN) and 1 within factor (time; Pre- vs. Post- test). Multiple comparisons were made with the Tukey's honestly significant difference (HSD) post hoc test when the Greenhouse-Geisser epsilon correction factor was >0.50, or

with the Bonferroni post hoc test when the epsilon was <0.05 . All analyses were made using Sigmaplot 11.0 software (Systat Software, Inc., San Jose, CA, USA). Significance level was set at $p \leq 0.05$.

In addition, an approach based on the magnitudes of differences (27) was used to assess practical significance. Time as well as condition differences were expressed as standardized difference or Cohen's d ($\pm 95\%$ confidence limits) using pooled SD . Threshold values for Cohen's d statistics were >0.2 – 0.5 (small), >0.5 – 0.8 (moderate), and >0.8 (large). To compare between conditions changes (i.e., period and training effect factors), the chance that the true (unknown) changes for Post- or RSH were higher (i.e., higher than the smallest practically important difference, or the smallest worthwhile [difference] change [0.2 multiplied by the between-subject SD , based on Cohen's d principle]), similar or lower were calculated. Quantitative chances of greater or lower values were assessed qualitatively as follows: $<1\%$, almost certainly not; 1 – 5% , very unlikely; 5 – 25% , unlikely; 25 – 75% , possible; 75 – 95% , likely; 95 – 99% , very likely; $>99\%$, almost certain. If the chances of having higher or lower values were both $>5\%$, the true difference was assessed as unclear (27).

RESULTS

Normoxic and Hypoxic Training Exposure

All players were able to combine their usual football-specific training sessions with the experimental-specific training sessions throughout the duration of the study. Number of training sessions, volume, and intensity were identically matched among the 2 groups. During training sessions over the 5-week training period, averaged SpO_2 values of the RSH group decreased at 91 – 93% , whereas it remained at 97 – 98% for the RSN group.

Performance

From Pre- to Post- training, the lower-limb explosive power (CMJ) improved ($p < 0.001$) to a similar extent in both groups ($+6.5 \pm 1.9\%$ vs. $+5.0 \pm 7.6\%$ for RSH and RSN, respectively). However, benefit appeared most likely for RSH, whereas it was only possible for RSN with a possibly beneficial effect for RSH compared with RSN (Table 4). Although sprinting performances increased significantly after 5 weeks of training compared with baseline (Pre-) with no significant difference between groups (Table 4), greater magnitudes were observed for RSH for all sprinting distances (likely to most likely beneficial) than that for RSN (possibly to likely beneficial). Qualitative inferences between groups appeared possibly beneficial for all sprint distances except for 10-m sprint (likely beneficial) for RSH compared with RSN. Despite MAS being unchanged throughout the protocol ($+0.9 \pm 2.3\%$ and $-0.2 \pm 4.6\%$ for RSH and RSN, respectively), there was, however, a greater magnitude perceived for RSH (possibly beneficial) than that for RSN (unlikely beneficial) with a possible beneficial effect for RSH compared with RSN (Table 4).

Both RSH and RSN groups improved RSA_{best} ($-3.0 \pm 1.7\%$ and $-2.3 \pm 1.8\%$, respectively; both $p \leq 0.05$) and RSA_{mean} ($-3.2 \pm 1.7\%$, $p < 0.01$ and $-1.9 \pm 2.6\%$, $p \leq 0.05$ for RSH and RSN, respectively; Figures 1A, B), whereas RSA_{Sdec} did not change ($-0.5 \pm 0.6\%$ and $-0.1 \pm 1.1\%$ for RSH and RSN, respectively; Figure 1C). There was no significant interaction between time and condition for any RSA-related parameter. Although training impact appeared of similar magnitude on RSA_{best} and RSA_{mean} (ranging from very likely to most likely) for both groups, RSH seemed possibly beneficial compared with RSN (Table 5). The RA_{Sdec} remained unchanged between Pre- and Post- in both groups ($-0.1 \pm 1.9\%$ and $-0.7 \pm 0.9\%$ for RSH and RSN, respectively; Figure 2C). Significant condition $\times$ time interactions effects were found for RA_{best} ($p \leq 0.05$) and RA_{mean} ($p \leq 0.05$) with larger decrease for RSH ($-4.4 \pm 1.9\%$ and $-4.3 \pm 0.6\%$ for RA_{best} and RA_{mean} , respectively; $p < 0.001$) than that for RSN ($-2.0 \pm 1.7\%$ and $-2.4 \pm 1.7\%$, respectively; $p \leq 0.05$) group (Figures 2A, B). Repeated-agility performance-related parameters improved in both condition with a very likely advantage for RSH compared with RSN (Table 5).

DISCUSSION

In the present study, we investigated the effect of hypoxia exposure when 2 weekly high-intensity intermittent running-based training sessions were added over a 5-week in-season period to the usual football training routine of highly-trained under-18 male players. **The major findings of this investigation were that (a) the addition of 10 specific run-based training sessions to their regular football practice substantially improved several neuromuscular fitness components related to on-field football physical performance, and (b) high-intensity training in normobaric hypoxia in a team sport applied setting (i.e., high-intensity intermittent and repeated-sprint runs, agility, and conditioning exercises) appeared more efficient than the same training in normoxia at enhancing RA ability, with the inclusion of direction changes.**

The addition of high-intensity intermittent running, sprints, and all-out efforts into normoxic training programs has been shown to be effective for team sport players (9,10). It is therefore believed that these types of training (e.g., repeated-sprint training or sprint interval training) challenge at different respective levels relative to the training content, both the metabolic and the neuromuscular/musculoskeletal systems (8). However, despite our limited understanding of these dose-response relationships between the training load and training-induced changes in physical capacities and performance, the primarily expected benefits of such exercises are to maximize cardiorespiratory fitness (10). The results of our study did not show any significant improvement in MAS for either RSH or RSN groups, although RSH was found possibly more effective on MAS than RSN. This was expected because the total duration of hypoxic exposure (e.g., hypoxic

dose) was too low for inducing any positive hematological adaptations. Moreover, intensities and exercise-rest ratios probably did not elicit $\dot{V}O_2$ responses near $\dot{V}O_{2\max}$ (38,41) and normoxic $\dot{V}O_{2\max}$ or peak power output are not further enhanced by hypoxic than normoxic training (52). Rather, other central (i.e., ventilatory, hemodynamics, or neural adaptations) or peripheral (i.e., muscle-buffering capacity, economy, mitochondrial biogenesis, lactate transport, pH regulation) factors outside hematological adaptations have also the potential to improve match-related performance in team sports (19). Because team sport training implicates a myriad of metabolic and neuromuscular systems simultaneously (26), anaerobic glycolytic energy contribution and neuromuscular load/musculoskeletal strain are logically likely the more important variables to consider (9).

During high-intensity intermittent exercise, exercise capacity not only depends on energy supply and energy depletion but also on the function of the neuromechanical system (9). Resistance exercise training is well known to elicit morphological and functional adaptations in the skeletal muscle (42). Degrees of muscular hypertrophy and strength gains after resistance training are thought to be dependent on the intensity of exercise, in such a way that an intensity of approximately 70–85% of 1 repetition maximum (1RM) is required to achieve a substantial effect (36). However, recent research suggests that similar muscle mass and strength gains can be accommodated with low or high-intensity resistance exercise training performed under intermittent hypoxic conditions (33,44). Nishimura et al. (44) reported accelerated increases in muscle cross-sectional area of the elbow flexors and extensors in untrained male student after 6 weeks of hypoxic resistance training (4 sets of 10 repetitions at 70% 1RM, 2 times per week, $F_{I}O_2 = 16\%$) compared with normoxia. In the present study, the improvement in lower-limb explosive strength (6.5%) and sprinting performances (6.0% at 10 m) following RSH indicate greater magnitude (possibly and likely higher benefits) compared with RSN improvement (5.0 and 4.3%, respectively). These greater hypoxic-induced strength gains might at least partially explain the larger improvements for agility factors (~4% for RSH vs. ~2% for RSN), which arguably request higher muscular solicitation (e.g., eccentric load during the braking phase of COD). Such improvements could not be observed during our RSA test, as the strength component is less decisive on this test where sprints are performed in the same direction, presumably resulting in an absence of difference between conditions. It is possible that this improved RA performance (i.e., as a result of a better production/force application technique) is because of neural adaptations. Although speculative, this may include increased motor unit synchronization or agonist muscle activation, as evidenced from greater electromyography signal amplitudes during posthypoxic intervention maximal voluntary contractions (33).

Because football efforts are characterized by high-intensity running bouts repeated throughout the game, improving

RSA is important to delay premature or excessive fatigue, eventually improving players' match-related physical performance (48). Our results, showing substantial improvements of both RSA (Figure 1) and RA (Figure 2), therefore confirm the practical relevance of the present training intervention. However, the question of the transfer of such training-induced individual physical performance improvements (as inferred from specific performance tests) to match performance enhancement (i.e., high-intensity running or involvement with the ball) and ultimately team's game results is highly debated and unknown (16,17). It is particularly interesting that RA was further improved after RSH as, in addition to the ability to repeat high-intensity efforts, the ability to COD is an important fitness component for successful participation in football (51). Introducing COD into repeated-sprint sequences increases the mechanical demands of the repeated accelerations inherent to consecutive COD (46). In fact, such running modality is likely to increase peripheral (particularly biarticular locomotor muscles) demands, resulting in a greater taxing of the anaerobic energy system (1). Consecutive training adaptation would be partly related to an upregulation of the glycolytic potential and to an increased anaerobic capacity (33,35), which might be helpful for team sport athletes. Besides, by challenging the functional reserve in muscle oxygen diffusing capacity likely used in hypoxia (15), RSH has the potential to stimulate beneficial adaptations in terms of phosphocreatine resynthesis and oxygen utilization mediated by hypoxia-inducible factors at the muscular level. Based on the suggestion that (a) the ability to resynthesize phosphocreatine is probably the major determinant of RSA (37), and (b) RSH efficiency is likely to be fiber-type dependent (19), adding a hypoxic stimulus to training can maximize fatigue resistance during high-intensity intermittent exercises.

One potential limitation of this study is the omission of resting skeletal muscle microbiopsy samples to confirm putative training-induced adaptive mechanisms at the muscular level (20). The main reason not to perform invasive measurement belongs to ethical considerations as this study was conducted within a realistic applied setting on developing athletes. Besides, the failure of previous researches to observe any significant change in total hemoglobin mass, red cell volume, or any other red cell indices after intermittent hypoxic training compared with sea level (2,24) convinced us not to investigate erythropoiesis response here because the low "altitude dose" associated with is unlikely to stimulate "the erythropoietic pathway to the point that it enhance postaltitude sea-level endurance performance" (32,56). Another point to mention is that performance changes should not only be monitored shortly (i.e., few days) after the intervention but also few weeks after the last day of exposure to distinguish the short (or immediate) from mid/long-term (or delayed) effects. Indeed, after return to sea level, performance might depend from some or all of the individual time course of

the changes in red blood cell mass, ventilatory acclimatization, biomechanical, and neuromuscular factors (18). Last, it is worth noting that whereas statistical significance and standardized magnitude of differences indicate greater increase for RSH than for RSN, caution is needed in interpreting our results given the large CI reported. Moreover, despite the double-blinded design used in this study, the addition of a “control” group (i.e., without the 2 weekly high-intensity sessions) to assess the fitness improvement of such training is lacking.

In conclusion, the use of the so-called RSH, a new hypoxic training method developed to overcome some of the inherent limitations of intermittent hypoxic training was shown to be as efficient as RSN for lower-limb explosive power and maximal sprinting performance improvement and to improve RSA performance of highly trained under-18 male footballers. A key feature of this study that used a carefully controlled double-blind design in a team sport-applied setting over a 5-week period was to demonstrate that the addition of 10 RSH sessions to players’ regular football training induced twice larger enhancements of RA ability than the same training in normoxia. Although RSH is a promising training strategy to induce a synergetic effect on performance over sea-level training in young high-level football players, further applied research bringing together physiologists and conditioning coaches is still needed to improve our understanding of how optimally design exercise and recovery parameters (intensities or exercise-rest ratios (39)) when implementing an RSH training regimen with team sports players.

PRACTICAL APPLICATIONS

This study reported that football-specific high-intensity and repeated-sprint exercises, be it performed in normoxic or hypoxic condition, is efficient to improve physical fitness performance factors in young football players. Such training intervention should therefore be recommended during preparation or in-season to boost players’ performance and delay premature fatigue. Moreover, performing repeated-sprint and RA sequences at maximal intensity in hypoxia may provide additional activation of anaerobic and neuromuscular pathways beyond that observed in normoxia. Therefore, the higher practical effect of RSH shown in this study suggests that the prescription of high-intensity hypoxic exercises is sufficient to have a synergetic effect on physical performance over sea-level training in football players. As also supported by recent evidence (22), RSH may also lead to putative benefits for other team sports like, rugby union or field hockey, where the ability to repeat high speed runs during an entire game is essential for overall performance.

Given the relevance of explosive power for footballers and the prevalence of neuromuscular qualities determining RSA (5), the present results suggest that adding hypoxic explosive exercises (i.e., heavy resistance or plyometric/agility [i.e.,

COD]/sprint drills) improve power-related factors and fatigue resistance. However, it is worth noting that no uniform gains resulting from an hypoxic training intervention should be expected for all age, level, and gender, because of (a) possible individual responsiveness (adherence, compliance, and variability) to hypoxia and (b) position and tactical dependence to match running performance (40). Besides, in an applied setting, the benefits-to-costs ratio and the use of this method have to be weighted with factors such as practicalities, financial, and logistics of this type of approach and the magnitude of the transfer of the RSA and RA improvements to match performance (e.g., high-intensity running and involvements with the ball during competitions) (2).

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